

Jacob Berger

Corvallis, Oregon | (971) 279-3048 | jacob@bergerj.com | bergerj.com

EDUCATION

Oregon State University — Corvallis, OR

Ph.D. & M.S. in Artificial Intelligence

Expected June 2030

Advisor: Dr. Lizhong Chen, STAR Lab. Graduate coursework: Deep Learning, Parallel Programming, Numerical ODEs/PDEs, Math of Data Science.

Honors B.S. in Applied Computer Science & Computational Mathematics

Sept 2022 – June 2026 • Summa Cum Laude (both degrees) • GPA: 3.9+

Accelerated Master's Program (graduate coursework as undergraduate). Honors degree. AI concentration, double major.

RESEARCH & PUBLICATIONS

Honors Thesis: Semantic Constraints in NLP

June 2025 – Present

Researcher (Advisor: Dr. Lizhong Chen, STAR Lab)

- First-author submission under review at AAAI 2027 ("Parsimonious Decoding"). Developed novel metrics to enforce semantic constraints in probabilistic language models, improving vocabulary legality without degrading fluency.

EXPERIENCE

AI Research Engineer

March 2023 – Present

Lexel Synergetics — Portland, OR

- Co-authored and secured a Phase I SBIR grant (federal R&D funding) for NLP-based language model development.
- Architected end-to-end data pipelines for training and fine-tuning specialized large language models (LoRA, RAG).
- Deployed cloud infrastructure on AWS, GCP, and Vercel for model training and serving.

Senior Capstone: AI for Home Automation

Sept 2025 – June 2026

Project Team Member — Corvallis, OR

- Integrated LLMs with Home Assistant for natural-language home automation. Presented at OSU Engineering Research Expo.

Head Mechanic

Dec 2022 – Present

MU Lanes and Games — Corvallis, OR

- Manage maintenance and diagnostics for electromechanical bowling lane systems.

PROJECTS

AuTuber — AI Stage Hand for Live Streaming

- LLM-driven agent observing webcam, screen, and audio to autonomously control OBS and VTube Studio via WebSocket. Team of 4, served as PM and AI engineer.

RelCube — Deep RL Rubik's Cube Solver

- Transformer-based solver (~3.5M params) using structured piece representation. Trained with Approximate Value Iteration and Expanding Curriculum. Team of 3.

SKILLS

Languages: Python, C++, Java, JavaScript, Bash, x86 Assembly, MATLAB

Machine Learning: PyTorch, HuggingFace Transformers, LLM Fine-tuning (LoRA), RAG Pipelines, Reinforcement Learning

Systems & Cloud: Linux (Debian), Proxmox VE, Virtualization (LXC/KVM), AWS (EC2/S3), GCP, Git, Vercel

Human Languages: Fluent Spanish (9 years immersion)

AWARDS

- Putnam Math Competition: Top 3 finisher at OSU.
- Finley Academic Excellence Scholarship: \$20,000 recipient.

EXTRACURRICULARS

- Competitive Math: OSU Putnam and COMAP Math Modeling (implemented Cellular Automata for ecosystem resilience analysis).
- Competitive Sports: Founded OSU national collegiate bowling team (All-Star qualifier); Rose City Rowing Club leader (2-time national qualifier).